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| **Status** | Finished |
| **Started** | Wednesday, 21 October 2026, 1:40 PM |
| **Completed** | Wednesday, 21 October 2026, 1:43 PM |
| **Duration** | 3 mins 36 secs |
| **Grade** | **15.00** out of 15.00 (**100**%) |

Question **1**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is small and sorted
c. When the list is small and unsorted ✓
d. When the list is large and unsorted

Question **2**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)
b. Binary ✓
c. Hash
d. Linear

Question **3**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Linear search ✓
b. Both (1) & (3)
c. Hash search
d. Binary search

Question **4**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is somewhere in the middle of the array ✓
- b. When the item is not the array at all
- c. Item is the last element in the array or item is not there at all
- d. When the item is the last element in the array

Question **5**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O👎
- b. O(1) ✓
- c. O(n log n)
- d. O(log n)

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. There must be a mechanism to delete and/or insert elements in the list ✓
- b. There should be direct access to the middle element in any sublist
- c. The list must be sorted
- d. Number values should only be present

Number values should only be present

Question **7**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Binary search
- b. Linear search ✓
- c. Hash search
- d. Interpolation search

Question **8**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Sequential search
- b. Binary search ✓
- c. Linear search
- d. Hash search

Question **9**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n/2
- b. n ✓
- c. n-1
- d. log n

Question **10**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94

b. 89 and 99

c. 90 and 99 ✓

d. 89 and 94

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly

b. Binary search algorithm is not efficient when the data elements more than 1500 ✓

c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

d. Must use a sorted array

Question **12**

Correct

Mark 1.00 out of 1.00

**_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.**

a. Complexity

b. Selection

c. Bubble ✓

d. Insertion

Question **13**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 500
c. 1000
d. 10 ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)
b. O👎 ✓
c. O(n log n)
d. O(1)

Question **15**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Sorting elements in a list
b. Inserting elements into a list
c. Deleting elements from a list
d. Determining whether an element is present in a list ✓